

CS & IT ENGINEERING



Operating System

Memory Management

Lecture – 3

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Recap of Previous Lecture



Topic

Contiguous Memory Management Technique

Topic

External & Internal Fragmentation

Topics to be Covered



Topic

Non-Contiguous Memory Mgmt. Technique

Topic

Paging

#Q. Consider a fixed partition MMT where there are 5 partitions of size 100MB, 250MB, 200MB, 500MB and 300MB. All Partitions are initially empty. The following process requests are made in the given order:

Process	Size
P1	150MB
P2	400MB
P3	270MB
P4	180MB
P5	80MB

	D.O.M.	Internal Frag.
first	5	270 MB
Best	5	270 MB
worst	4	570 MB

Provide the following answers for First fit, Best fit and Worst Fit policies?

Maximum degree of multiprogramming?

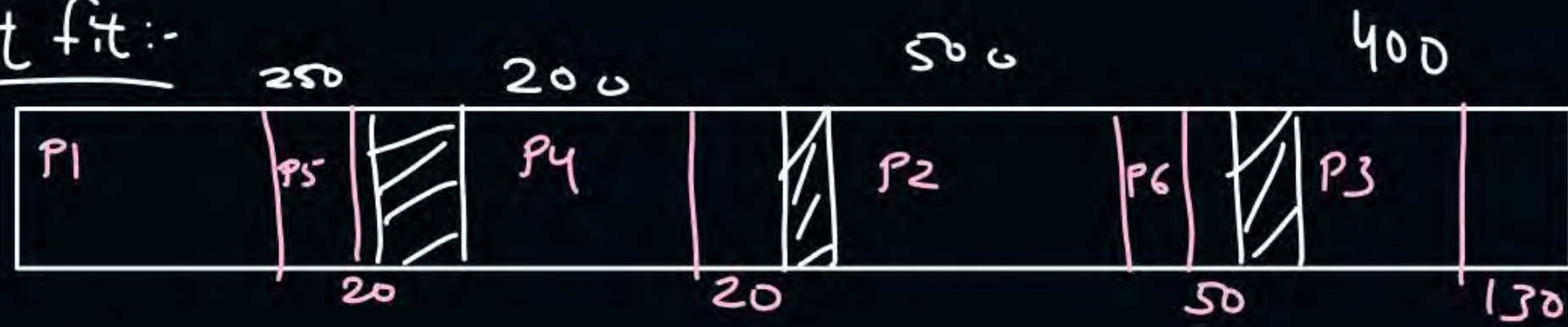
What is the total internal fragmentation size?

#Q. Consider variable partition MMT where there are 4 partitions of size 250MB, 200MB, 500MB and 400MB. The following process requests are made in the given order:

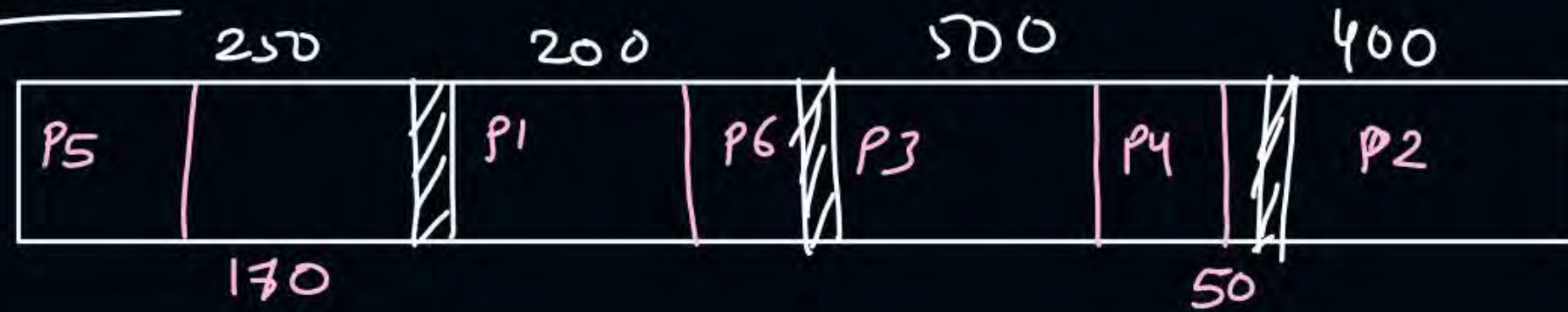
Process	Size
P1	150MB
P2	400MB
P3	270MB
P4	180MB
P5	80MB
P6	50MB

Provide how the processes are stored for First fit, Best fit and Worst Fit policies?

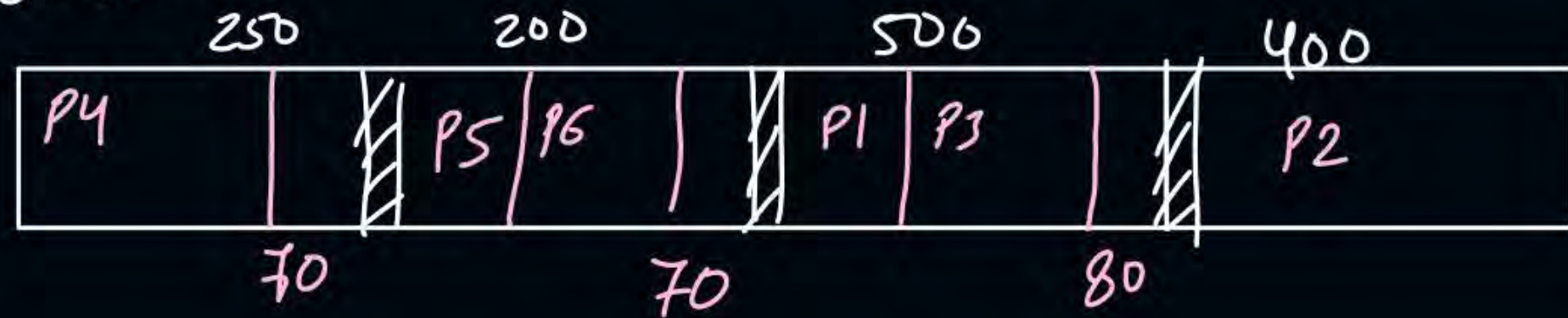
first fit:-



Best fit:-



worst fit



Non-Contiguous mem. mgmt. Technique:-

Process is divided into parts & each part can be stored any where in the mem.

Types

Paging

Process is divided into equal size partitions

Segmentation

Process is divided into variable size partitions



Topic : Paging



- Process is divided in equal size of pages
- Physical memory is divided in same equal size of frames
- Pages are scattered in frames



Topic : Paging

Example: Process $\Rightarrow$ 4 Pages
m.m. $\Rightarrow$ 8 frames

Process		Page Table	
Page 0		0	4
Page 1		1	2
Page 2		2	7
Page 3		3	1

m.m. (Physical mem.)

	Frame 0
Page 3	Frame 1
Page 1	2
	3
Page 0	4
	5
	6
Page 2	7



Topic : Paging

Consider

- A process has 4 pages
- Main memory has 8 frames

2 bits

3 bits

00	Page 00
01	Page 01
10	10
11	11

00	100
01	010
10	111
11	001

Page table

000	
001	Page 11
010	Page 01
011	
100	Page 00
101	
110	
111	Page 10

Assume page size = 2 bytes

$$\text{Process size} = \text{No. of pages} * 1 \text{ page size}$$

$$= 4 * 2B$$

$$= 8B \Rightarrow 000 \text{ to } 111 \text{ (0 to 7)}$$

$$\text{mm size} = \text{No. of frames} * 1 \text{ frame size}$$

$$= 8 * 2B$$

$$= 16B \Rightarrow \text{byte no.} = 0 \text{ to } 15 \\ = 0000 \text{ to } 1111$$

$$\text{No. of pages in process} = \frac{\text{Process size}}{\text{Page size}}$$

$$\text{No. of frames} = \frac{\text{Physical mem. size}}{\text{Page (frame) size}}$$



Topic : Paging

Page number

Process			Page Table	
000	a	00	00	100
001	b			
010	c	01	01	010
011	d			
100	e	10	10	111
101	f			
110	g	11	11	001
111	h			

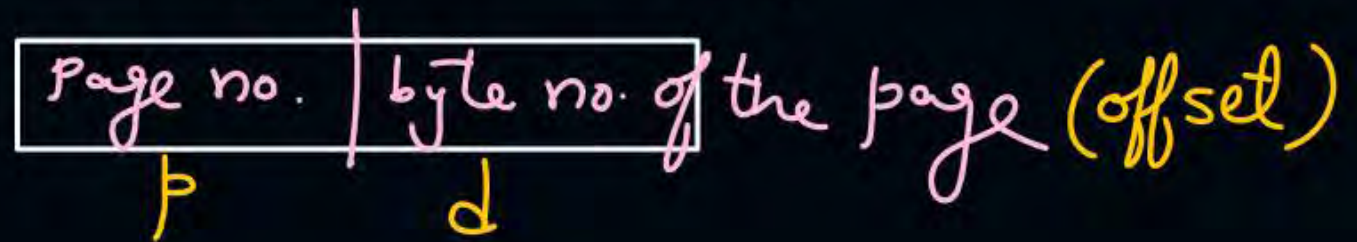
logical addresses

Physical mem. addresses

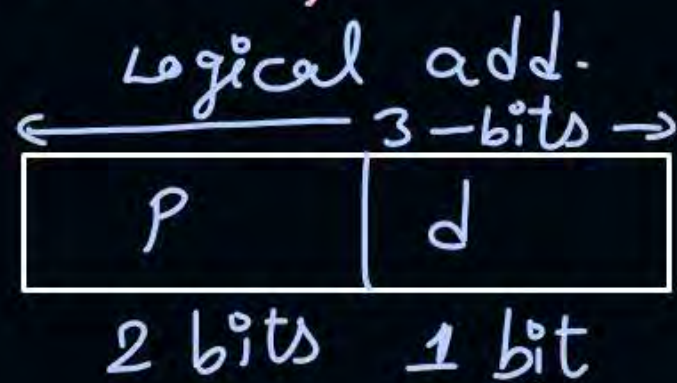
Physical Memory

0000		frame 000
0001		
0010	g	001
0011	h	
0100	c	010
0101	d	
0110		011
0111		
1000	a	100
1001	b	
1010		101
1011		
1100		110
1101		
1110	e	111
1111	f	

Logical address



In our example



no. of bits needed for offset = $\log_2(\text{Page size in bytes})$

no. of bits needed for page no. = $\log_2(\text{no. of pages in process})$

page size	bits for d
4 bytes = 2^2 B	2 bits
8 bytes = 2^3 B	3 bits
32 B = 2^5 B	5 bits
2 KB = 2^{11} B	11 bits
16 KB = 2^{14} B	14 bits

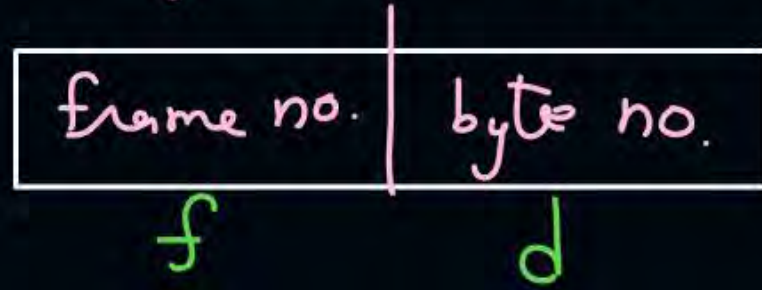
$$k = 2^{10}$$

$$m = 2^{20}$$

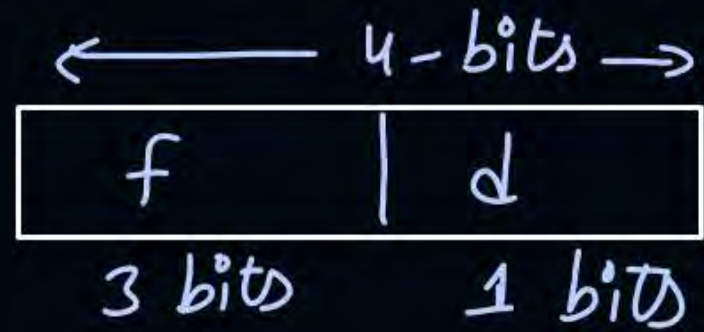
$$n = 2^{30}$$

No. of pages	bits for p
2^{16}	16 bits
2^{30}	30 bits
2^6	6 bits

physical address



for our example:-



CPU generates logical address $\Rightarrow (011)_2$

P	d
01	1

search
into page table

frame no.
010

f	d
010	1

physical add $= (0101)_2$



Topic : Paging



CPU always generates
logical address

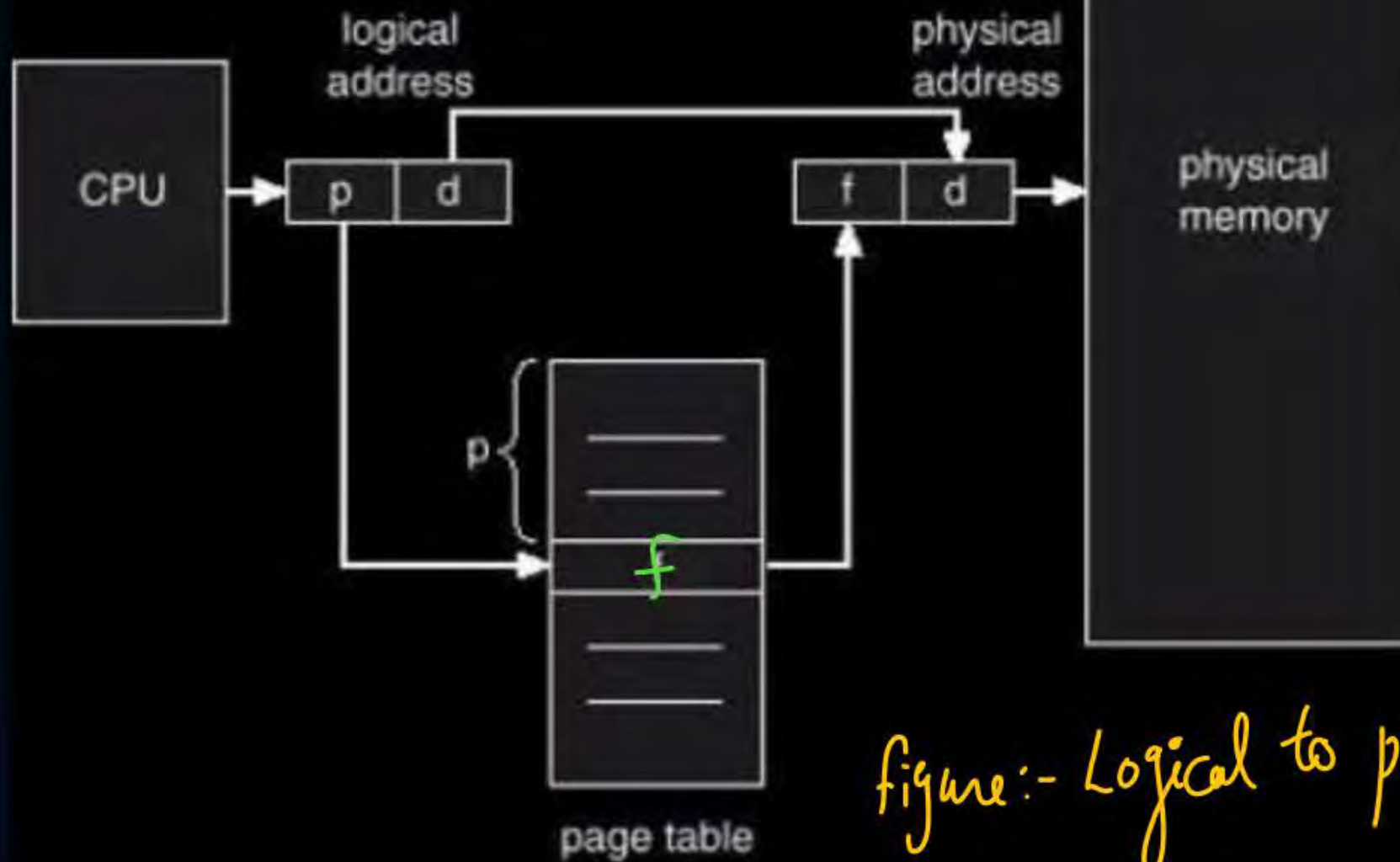


figure:- Logical to physical address translation



Topic : Paging

- Process is divided in equal size of pages
- Physical memory is divided in same equal size of frames
- Processor will have a view of process and its pages
- Pages are scattered in frames
- Page table is used to map a process page to a physical frame
- Number of entries in page table = Number of pages in process
- OS maintains a page table for each process $\rightarrow$ for protection

Logical address space $\Rightarrow$ process size
(L.A.S.)

physical $\text{---} || \text{---} \Rightarrow$ main mem. size
(P.A.S.)

bits for Logical address = $\log_2$ (L.A.S. in bytes)

bits for physical address = $\log_2$ (P.A.S. in bytes)



2 mins Summary

Topic

Non-Contiguous Memory Mgmt. Technique

Topic

Paging



Happy Learning

THANK - YOU